

Cada tipo de sensor genera un JSON con:

- **sensor_id**: Identificador único del sensor.
- **sensor_type**: Tipo de sensor (weather, traffic_counter, traffic_light, information_display).
- **street_id**: Calle asociada al sensor.
- **timestamp**: Marca de tiempo del dato generado.
- **location**: Información de ubicación (latitud, longitud, distrito, barrio, etc.).
- **data**: Datos específicos del sensor, que varían según el tipo.

Weather_stations

```
{  
  "sensor_id": "WS_001",  
  "sensor_type": "weather",  
  "street_id": "ST_001",  
  "timestamp": "2025-10-15T12:34:56.789Z",  
  "location": {  
    "latitude": 40.416775,  
    "longitude": -3.703790,  
    "altitude_meters": 650.0,  
    "district": "Centro",  
    "neighborhood": "Sol"  
  },  
  "data": {  
    "temperature_celsius": 22.5,  
    "humidity_percent": 65.0,  
    "air_quality_index": 45,  
  }  
}
```

```
    "wind_speed_kmh": 12.3,  
    "wind_direction_degrees": 180,  
    "atmospheric_pressure_hpa": 1015.2,  
    "uv_index": 5  
  }  
}
```

traffic_counter

```
{  
  "sensor_id": "TC_001",  
  "sensor_type": "traffic_counter",  
  "street_id": "ST_002",  
  "timestamp": "2025-10-15T12:34:56.789Z",  
  "location": {  
    "latitude": 40.416775,  
    "longitude": -3.703790,  
    "district": "Centro",  
    "neighborhood": "Sol"  
  },  
  "data": {  
    "vehicle_count": 35,  
    "pedestrian_count": 120,  
    "bicycle_count": 5,  
    "direction": "north",  
    "counter_type": "vehicle",  
    "technology": "camera",  
    "average_speed_kmh": 28.5,  
  }  
}
```

```
    "occupancy_percentage": 75.0,  
    "traffic_density": "high"  
  }  
}
```

traffic_light

```
{  
  "sensor_id": "TL_001",  
  "sensor_type": "traffic_light",  
  "street_id": "ST_003",  
  "timestamp": "2025-10-15T12:34:56.789Z",  
  "location": {  
    "latitude": 40.416775,  
    "longitude": -3.703790,  
    "district": "Centro",  
    "neighborhood": "Sol"  
  },  
  "data": {  
    "current_state": "green",  
    "cycle_position_seconds": 15,  
    "time_remaining_seconds": 45,  
    "cycle_duration_seconds": 90,  
    "traffic_light_type": "mixed_vehicle_pedestrian",  
    "circulation_direction": "bidirectional",  
    "pedestrian_waiting": false,  
    "pedestrian_button_pressed": false,  
    "malfunction_detected": false,  
  }  
}
```

```
    "cycle_count": 3,  
    "state_changed": true,  
    "last_state_change": "2025-10-15T12:34:11.789Z"  
  }  
}
```

information_display

```
{  
  "sensor_id": "ID_001",  
  "sensor_type": "information_display",  
  "street_id": "ST_004",  
  "timestamp": "2025-10-15T12:34:56.789Z",  
  "location": {  
    "latitude": 40.416775,  
    "longitude": -3.703790,  
    "district": "Centro",  
    "neighborhood": "Sol"  
  },  
  "data": {  
    "display_status": "active",  
    "current_message": "Tráfico fluido",  
    "content_type": "traffic",  
    "brightness_level": 85,  
    "display_type": "led_matrix",  
    "display_size_inches": 420.0,  
    "supports_color": true,  
    "temperature_celsius": 45.3,  

```

"energy_consumption_watts": 250.5,

"last_content_update": "2025-10-15T12:34:56.789Z"

}

}